USB POWER / ALL-RAIL PROTECTION AND RESET

S6 schematic only. All three converters retained: main and + rails TPS16630: negative LM5067 + MOSFET.

USB PD and main eFuse

RUN
VPD
VIN
MAIN_OK

CH224A: 20 V request
TPS16630: 4.186 A nominal

File: 01_usb_input.kicad_sch

+12 V converter and output eFuse

MAIN_OK
P12_OK
RUN
V3V3
VIN
+12V
P12_RAW

TPS62933F -> TPS16630
+12 V protected output

File: 02_positive_12v.kicad_sch

+5 V converter and output eFuse

MAIN_OK
P5_OK
RUN
V3V3
VIN
+5V

TPS62933F -> TPS16630
+5 V protected output

File: 03_positive_5v.kicad_sch

Negative converter - raw rail

P12_RAW
VIN
N12_RAW

LMR38020F inverting buck-boost
RAW negative supply

File: 04_negative_converter.kicad_sch

Negative rail protection and level shifting

N12_RAW
NOT_RUN
V3V3
VIN
−12V
N12_STATUS

LM5067 + external N-MOSFET
1.0 A nominal / optical control

File: 05_negative_protection.kicad_sch

Protected outputs - voltage monitors and panel LEDs

+12V
+5V
−12V
MAIN_LED_OK
N12_STATUS
RUN
V3V3
VPD
P12_OK
P5_OK
N12_OK
V20_OK

Parallel Eurorack headers
MAIN, +12, +5 and −12 LEDs

File: 07_outputs_status.kicad_sch

Always-on control - all-rail fault latch and RESET

MAIN_OK
N12_OK
NOT_RUN
P12_OK
P5_OK
V20_OK
VPD
RUN
V3V3
MAIN_LED_OK

60 V always-on LDO / retained ARMED + FAULT
Manual RESET disables all three rails

File: 06_fault_reset.kicad_sch

FAULT POLICY
Each rail detects independently.
Persistent fault -> all OFF until RESET.
Main LED is not merely a +5 V lamp.

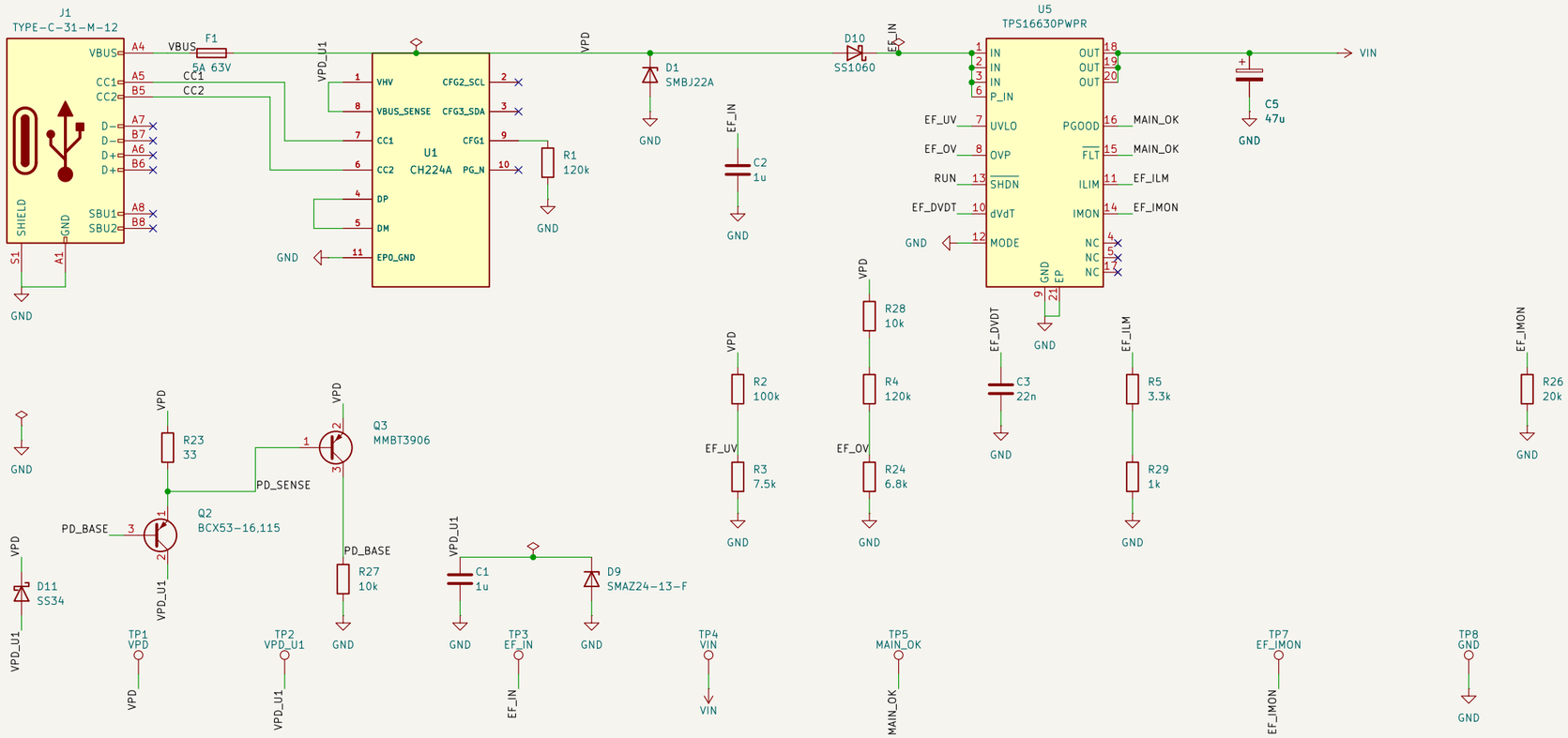
OUTPUTS: +12 / +5 / −12
Nominal limits: 2.647 / 2.647 / 1.0 A
Tolerance and thermal limits apply.
Not a guaranteed 54.5 W simultaneous rating.

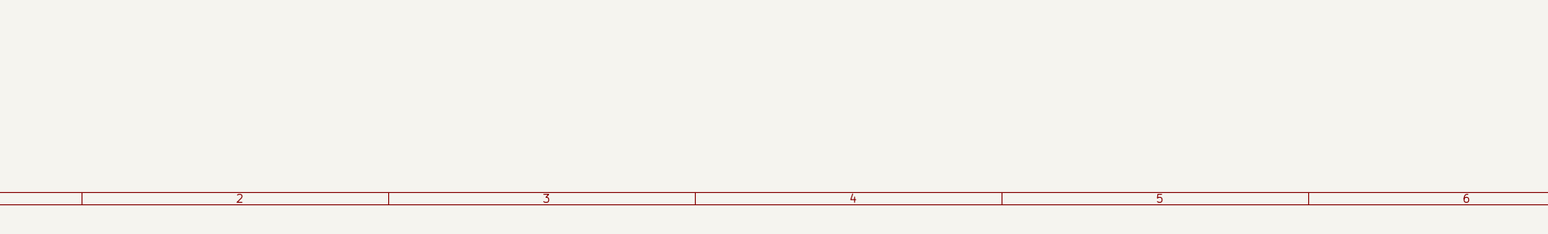
Design notes and budget: docs/design.md
S6: per-rail protection; all-rail fault latch/reset; four status LEDs
Big Modular
Sheet: /
File: usb-power-45w.kicad_sch

Title: USB power - wired power flow and protection overview

Size: User	Date:	**Rev: S6**
KiCad E.D.A. 10.0.6		Id: 1/8

USB PD AND MAIN EFUSE





20 V PD request: U5 TPS16630 remains 60 V, 4.186 A nominal. RUN from retained fault latch.
MAIN_OK is wired FLT + PGOOD at 3.3 V, NOT a raw-voltage logic output.

INTER-SHEET INTERFACE (see wired power-flow overview)

MAIN_OK ◇— MAIN_OK

RUN ◇— RUN

VIN ◇— VIN

VPD ◇— VPD

Schematic only — see design.md and s6-rail-protection.md

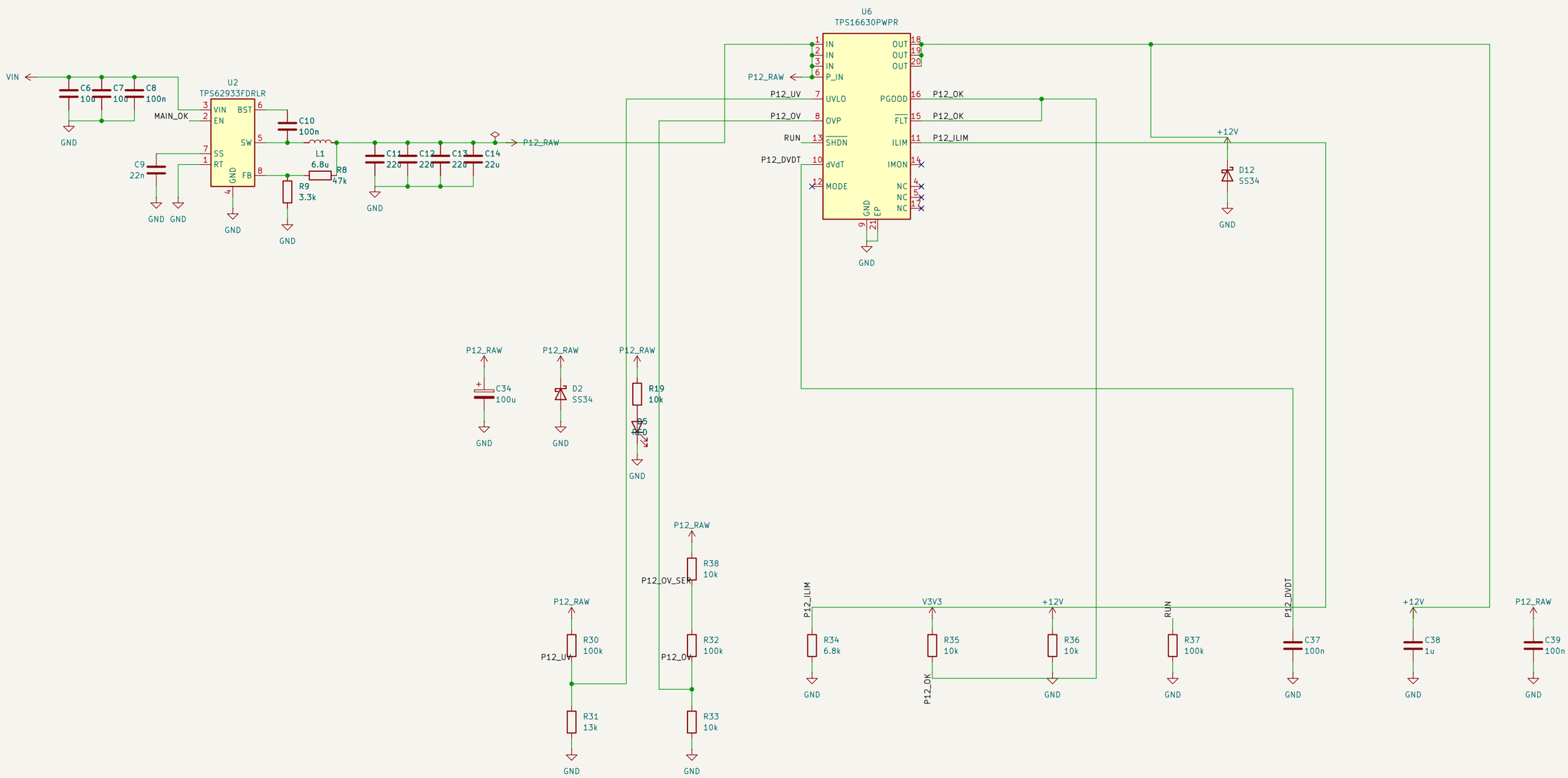
Big Modular

Sheet: /USB PD and main eFuse/
File: 01_usb_input.kicad_sch

Title: USB PD and main eFuse

Size: User	Date:	**Rev: S6**
KiCad E.D.A. 10.0.6		Id: 2/8

+12 V CONVERTER AND OUTPUT EFUSE



TPS62933F converter unchanged. RAW feedback/bulk stay BEFORE U6.
U6: 2.647 A nominal, 2.335–2.971 A screened. No guaranteed 2.5 A minimum; MODE latch-off.

INTER-SHEET INTERFACE (see wired power-flow overview)

+12V ◇ +12V

RUN ◇ RUN

MAIN_OK ◇ MAIN_OK

V3V3 ◇ V3V3

P12_OK ◇ P12_OK

VIN ◇ VIN

P12_RAW ◇ P12_RAW

Schematic only — see design.md and s6-rail-protection.md

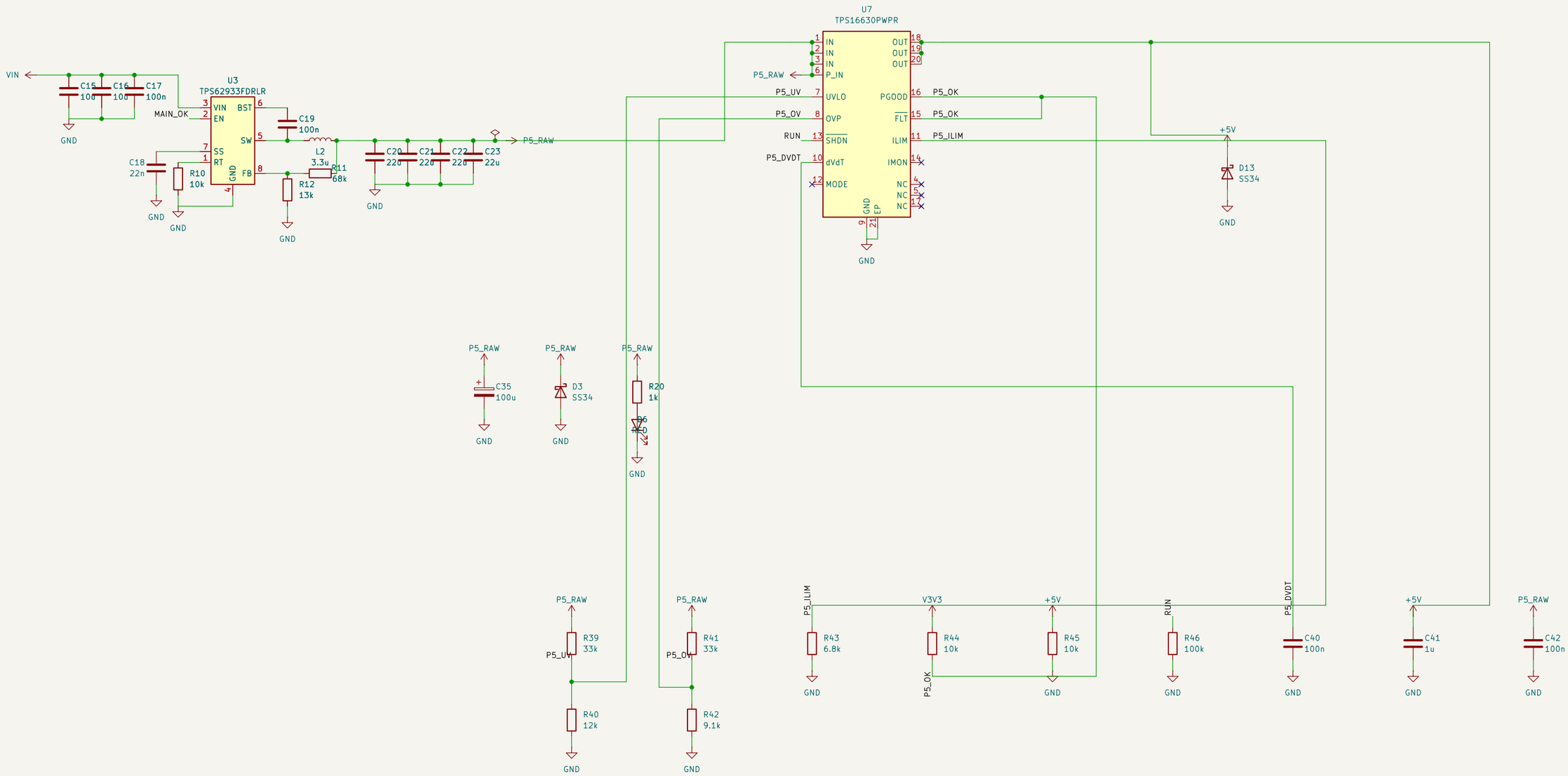
Big Modular

Sheet: /+12 V converter and output eFuse/
File: 02_positive_12v.kicad_sch

Title: +12 V converter and output eFuse

Size: User	Date:	**Rev: S6**
KiCad E.D.A. 10.0.6		Id: 3/8

+5 V CONVERTER AND OUTPUT EFUSE



TPS62933F converter unchanged. U7 input must remain >=4.5 V.
U7: 2.647 A nominal, 2.335–2.971 A screened. Output-valid and FLT/PGOOD share P5_OK.

INTER-SHEET INTERFACE (see wired power-flow overview)

+5V ◇ +5V

V3V3 ◇ V3V3

MAIN_OK ◇ MAIN_OK

VIN ◇ VIN

P5_OK ◇ P5_OK

RUN ◇ RUN

Schematic only — see design.md and s6-rail-protection.md

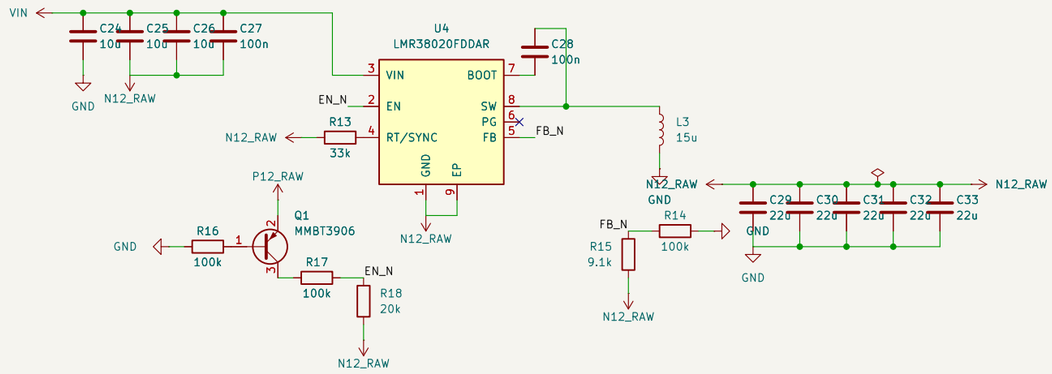
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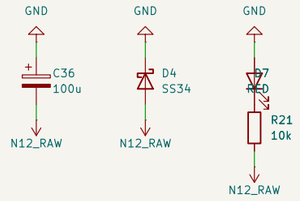
Sheet: /+5 V converter and output eFuse/
File: 03_positive_5v.kicad_sch

Title: +5 V converter and output eFuse

Size: User	Date:	**Rev: S6**
KiCad E.D.A. 10.0.6		Id: 4/8

NEGATIVE CONVERTER — RAW RAIL





LMR38020F ground/exposed pad = N12_RAW, not system GND.
Q1 follows P12_RAW: load reaches N12_RAW ONLY through page 05 series protection.

INTER-SHEET INTERFACE (see wired power-flow overview)

N12_RAW ◇ N12_RAW

P12_RAW ◇ P12_RAW

VIN ◇ VIN

Schematic only — see design.md and s6-rail-protection.md
Big Modular

Sheet: /Negative converter — raw rail/
File: 04_negative_converter.kicad_sch

Title: Negative converter — raw rail

Size: User	Date:	**Rev: S6**
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NEGATIVE RAIL PROTECTION AND LEVEL SHIFTING

System GND is continuous. R48 || R49 = 50 mOhm; Q4 is in the NEGATIVE conductor.
U8 VCC is resistor-fed from VIN. U16 disables via UVLO. U17 level shifts PGD.
Nominal 1 A / 3.1 W MOSFET limit. Reverse-collapse pin clamps do not block reverse power.

N_VCC

VIN
R50
10k
N_VCC

VIN
R51
10k
N_VCC

N_VCC
C43
1u
N12_RAW

U8
LM5067MM-2/NOPB
N_VCC 1 VCC
N_UV 2 UVLO
N_OV 3 OVLO
N_PWR 4 PWR
N12_RAW 5 VEE
N_TIMER 6 TIMER
SENSE 7 N_SENSE_PIN
GATE 8 N_GATE
OUT 9 N_OUT_PIN
PGD 10 N_PGD

N_SENSE
R89
33
N_SENSE_PIN
R90
10k
N_OUT_PIN

−12V
Q4
FQD13N06LTM
N_GATE
N_SENSE

N_SENSE
R48
0.1
N12_RAW

N_SENSE
R49
0.1
N12_RAW

V3V3
R59
1k
NEG_OFF_LED

V3V3
R60
1k
NEG_OFF_LED

U16
LTV-354T
NEG_OFF_LED
NEG_OFF_K
N_UV
N12_RAW

NEG_OFF_K
Q5
2N7002
NOT_RUN
GND

GND
R61
10k
N_PGD_LED

V3V3
R62
100k
N12_STATUS

U17
LTV-354T
N_PGD_LED
N_PGD
N12_STATUS
GND

GND
R52
68k
N_UV

GND
R54
47k
N_OV

N_PWR
R56
22k
N12_RAW

N_TIMER
C44
1u
N12_RAW

N_GATE
R57
100k
N_SENSE

R53
33k
N12_RAW

R55
10k
N12_RAW

GND
C45
1u
−12V

GND
R58
10k
−12V

BAT54SLT1G
N12_RAW
N_SENSE_PIN
D19
N_SENSE_PIN

BAT54SLT1G
N12_RAW
N_OUT_PIN
D20
N_OUT_PIN

GND
D14
SS34
−12V

−12V

INTER-SHEET INTERFACE (see wired power-flow overview)

−12V −12V
N12_RAW N12_RAW
N12_STATUS N12_STATUS
NOT_RUN NOT_RUN
V3V3 V3V3
VIN VIN

Schematic only — see design.md and s6-rail-protection.md
Big Modular
Sheet: /Negative rail protection and level shifting/
File: 05_negative_protection.kicad_sch
Title: Negative rail protection and level shifting

Size: User	Date:	**Rev: S6**
KiCad E.D.A. 10.0.6		Id: 6/8

ALWAYS-ON CONTROL — ALL-RAIL FAULT LATCH AND RESET

U9
TPS7A1633DGNR
C46 1u
C47 10u
R63 10k
MAIN_OK
RUN
R64 10k
U10
TPS3808G01DBVR
R91 15k
R65 47k
R66 10k
R67 10k
R68 10k
C48 100n
U11
TPS3808G01DBVR
R69 33k
R70 10k
C49 1u
R71 10k
C50 100n
SW1
TL3340AF160QG
RESET_BTN_N
U12
SN74HC74DR
U13
SN74HC08DR
U14
SN74HC20DR
R72 10k
C51 100n
U15
SN74LVC1G17DBVR
C52 100n
C53 100n
C54 100n
C55 100n
TP9 RUN
TP10 ARMED
TP11 FAULT_LATCH
TP12 P12_OK
TP13 P5_OK
TP14 N12_OK
TP15 V20_OK
TP16 V3V3
TP6 FAULT_LATCH

RESET clears ARMED and FAULT; RUN is forced low while reset is asserted.
Startup qualification ~5.7 s typical, then any persistent bad status latches OFF until RESET.
ARMED sheet interface (see wires & power symbols elsewhere). Local protection is active at startup.

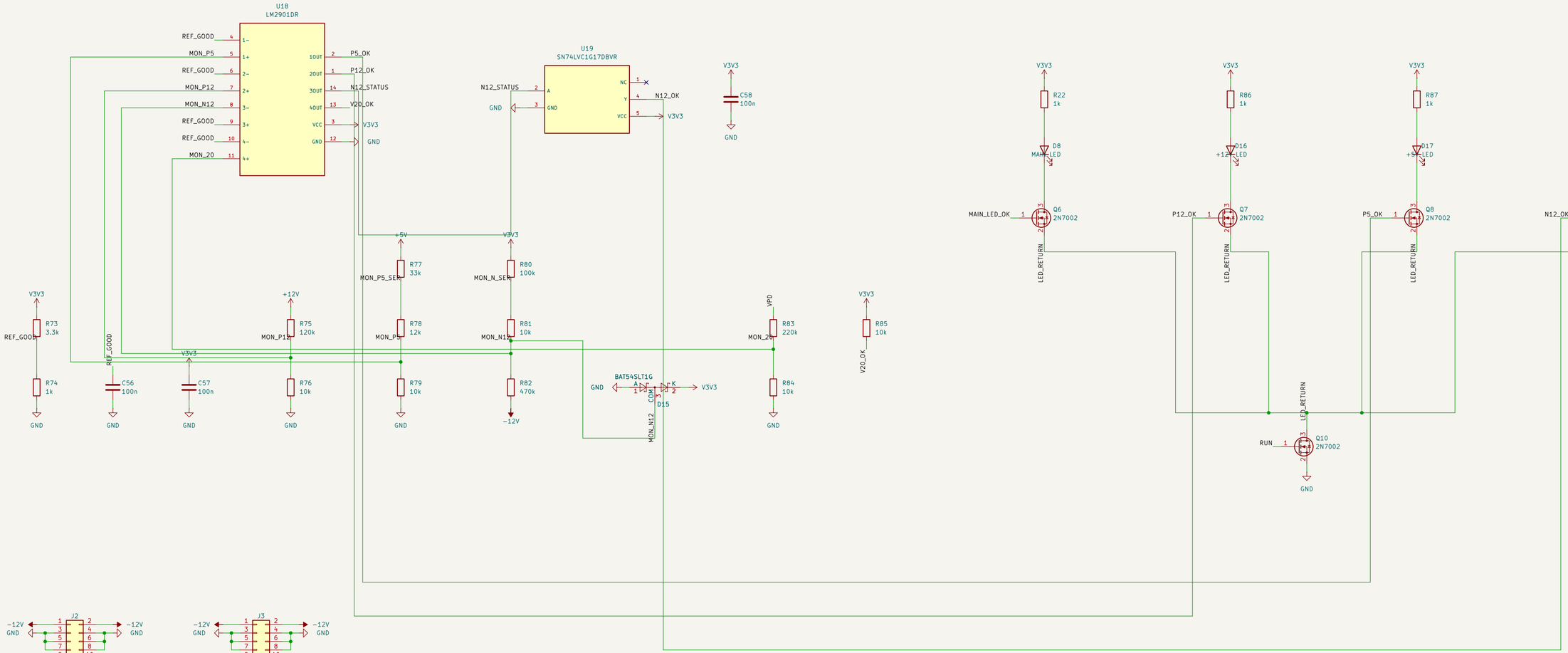
Schematic only — see design.md and s6-rail-protection.md
Big Modular
Sheet: /Always-on control — all-rail fault latch and RESET/
File: 06_fault_reset.kicad_sch

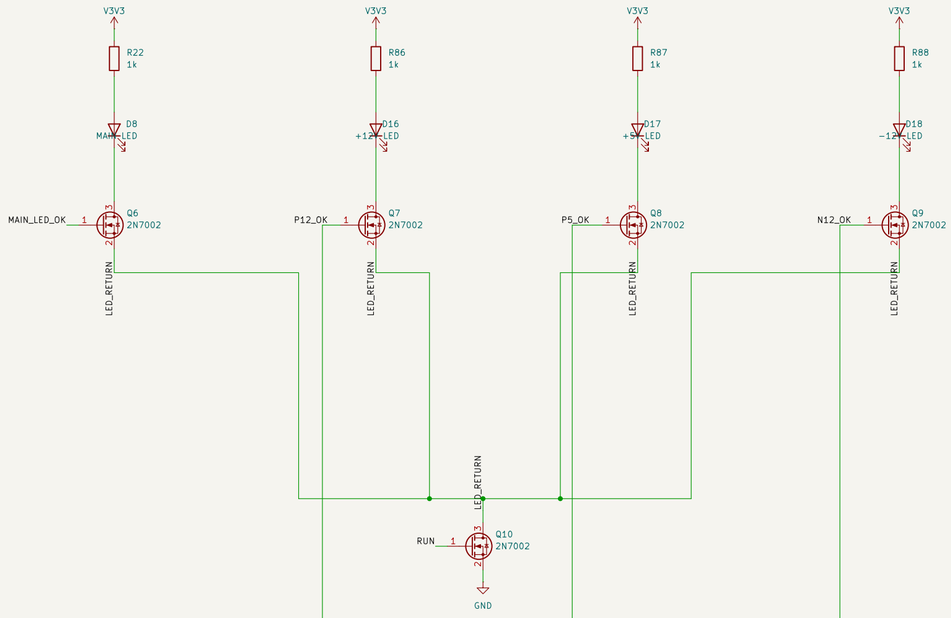
Title: Always-on control — all-rail fault latch and RESET

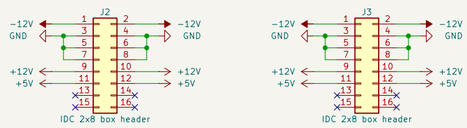
Size: User	Date:	**Rev: S6**
KiCad E.D.A. 10.0.6		Id: 7/8

PROTECTED OUTPUTS — VOLTAGE MONITORS AND PANEL LEDS

D8 MAIN: input-valid AND main-eFuse status AND RUN. D16/+12, D17/+5, D18/−12: delivered-rail status AND RUN.
Any latched common fault turns all four panel LEDs OFF. D5/D6/D7 on converter pages indicate RAW rails.
Voltage-present thresholds are not +/−5% regulation guarantees: retained external charge is not actively discharged.







INTER-SHEET INTERFACE (see wired power-flow overview)

+12V
N12_STATUS
V3V3
+5V
P12_OK
VPD
−12V
P5_OK
MAIN_LED_OK
RUN
N12_OK
V20_OK

Schematic only — see design.md and s6-rail-protection.md

Big Modular

Sheet: /Protected outputs — voltage monitors and panel LEDs/
File: 07_outputs_status.kicad_sch

Title: Protected outputs — voltage monitors and panel LEDs

Size: User	Date:	**Rev: S6**
KiCad E.D.A. 10.0.6		Id: 8/8